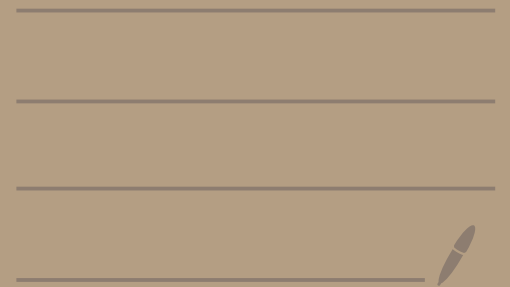
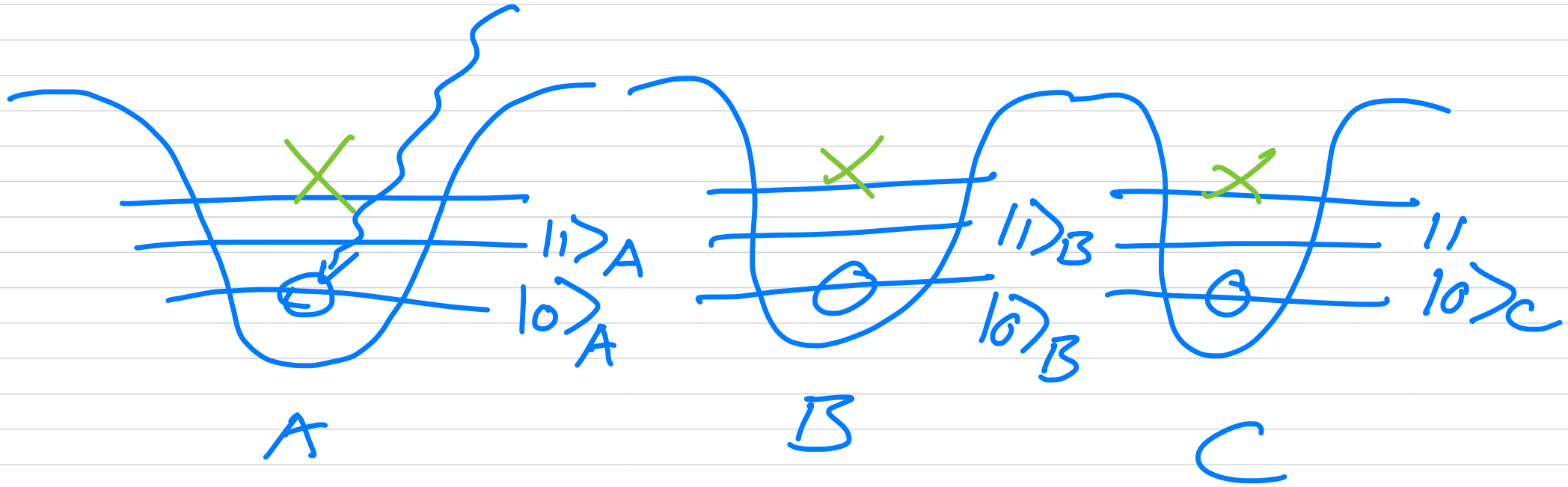


FYS5419/9419, lecture week 1,

January 21, 2025



FYS5419/9419 week 1



$$|0\rangle_A \otimes |0\rangle_B \otimes |0\rangle_C \rightarrow |000\rangle$$

(i) initialization

(ii) Time evolution

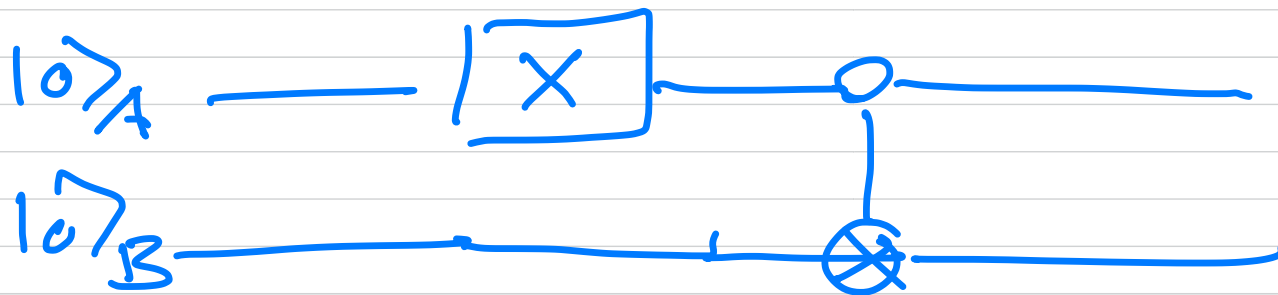
(iii) Measurement

(ii)

$|4\rangle_{\text{initial}}$

$$|4\rangle_{\text{final}} = U(t, t') |4\rangle_{\text{initial}}$$

t t'



$|4\rangle_{\text{final}}$

$$|0\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad \wedge \quad |1\rangle = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$\hat{P} = |0\rangle\langle 0| = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$$

$$\hat{Q} = |1\rangle\langle 1| = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$$

$$\hat{I} = \hat{P} + \hat{Q} = \sum_{i=0}^1 |i\rangle\langle i|$$

superposition

$$|\psi\rangle = \alpha |0\rangle + \beta |1\rangle$$

$$\hat{P}|\psi\rangle = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \left(\alpha \begin{bmatrix} 1 \\ 0 \end{bmatrix} + \beta \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right) = \alpha |0\rangle$$